

FOURTH ASSIGNMENT :

a. Even or Odd Checker : Task: Write a program that prompts the user for an integer. Use if/else to check whether the number is even or odd. Print "Even" or "Odd" accordingly.

```
In [ ]: a = int(input("Enter a number :"))
        if a % 2 == 0:
            print(f"{a} is even.")
        else:
            print(f"{a} is odd.")
```

12 is even.

b. Positive, Negative, or Zero : Task: Write a program that prompts the user for a number. Use if/elif/else to determine if the number is positive, negative, or zero. Print a message such as "The number is positive."

```
In [5]: b = int(input("Enter a number :"))
        if b < 0:
            print(f"{b} is negative.")
        elif b == 0:
            print(f"{b} is zero.")
        else:
            print(f"{b} is positive.")
```

13 is positive.

c. Grade Categorizer : Task: Prompt the user for a number between 0 and 100 (the score). Use if/elif/else to categorize the score: 90–100: "Grade A" 80–89: "Grade B" 70–79: "Grade C" < 70: "Fail"

```
In [7]: c = int(input("Enter your score : "))
        if c >= 90 and c <= 100:
            print(f"{c} is Grade A.")
        elif c >= 80 and c <= 89:
            print(f"{c} is Grade B.")
        elif c >= 70 and c <= 79:
            print(f"{c} is Grade C")
        else:
            print("Failed")
```

78 is Grade C

d. Simple Password Check : Task: Prompt the user for a password (e.g., "secret"). Use if to check if the user's input matches the correct password. If correct, print "Access granted", otherwise print "Access denied".

```
In [1]: password = input("Enter your password : ")
if password == "secret" or "Secret":
    print("Access granted.")
else:
    print("Access denied.")
```

Access granted.

e. Smallest of Three Numbers : Task: Prompt the user for three different numbers. Use if/elif/else to find and print which one is smallest.

```
In [ ]: num1 = int(input("Enter first number : "))
num2 = int(input("Enter second number : "))
num3 = int(input("Enter third number : "))
if (num1 >= num2 and num1 >= num3): # 12
    print(f"{num1} is greatest.")
elif(num2 >= num1 and num2 >= num3): # 13
    print(f"{num2} is greatest")
else:
    print(f"{num3} is greatest.") # 10
```

13 is greatest

f. Simple Menu Selection : Task: Prompt the user to choose an option (e.g., 1 for "Start", 2 for "Settings", 3 for "Exit"). Use if/elif/else to print a response based on which option is chosen. Example: If user enters 1: print "Game starting..." If user enters 2: print "Opening settings..." If user enters 3 or any other: print "Exiting..."

```
In [4]: print("Enter 1 for Start, 2 for Settings, and 3 for Exit.")
users = int(input("Enter a number:"))
if users == 1:
    print("Game Starting")
elif users == 2:
    print("Opening settings.")
else:
    print("Exiting")
```

Enter 1 for Start, 2 for Settings, and 3 for Exit.
Opening settings.